

Research Interests

Socially Responsible Operations, Fairness, Pricing, and Urban Systems

Education

- Aug. 2021 – **Ph.D. in Operations Research**, *Columbia University*, New York, USA
present Advisor: Prof. Adam N. Elmachtoub
- Sept. 2018 – **M.S. in Industrial & Systems Engineering**, *KAIST*, Daejeon, Korea
Aug. 2020 Advisor: Prof. Il-Chul Moon
- Feb. 2012 – **B.S. in Industrial & Systems Engineering and Electrical Engineering**, *KAIST*, Daejeon, Korea
Aug. 2018 Summa Cum Laude, Graduated first in the Department of Industrial & Systems Engineering

Publications

- [1] **Fair Aggregation in Virtual Power Plants**
Liudong Chen*, Hyemi Kim*, Adam N. Elmachtoub, Bolun Xu
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.
Submitted to Management Science
- [2] **Learning Fair Demand Models**
Adam N. Elmachtoub, Hyemi Kim, Jonathan Y. Tan (α - β)
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2026.
Submitted to Operations Research
- [3] **Fair Fares for Vehicle Sharing Systems**
Adam N. Elmachtoub and Hyemi Kim (α - β)
Operations Research, 2026.
(Non-Archival) *Conference on Fairness, Accountability, and Transparency (FAccT)*, 2025.
○ Finalist, *INFORMS DEI Best Student Paper Award*, 2024
○ Finalist, *INFORMS Transportation Science and Logistics Society Best Student Paper Award*, 2024
- [4] **Black-Box EM Algorithm for Estimating Latent States of High-Speed Vehicles**
Yoon-Yeong Kim, Hyemi Kim, WonSung Lee, Han-Lim Choi, and Il-Chul Moon
American Institute of Aeronautics and Astronautics, 2021.
- [5] **Counterfactual Fairness with Disentangled Causal Effect Variational Autoencoder**
Hyemi Kim, Seungjae Shin, JoonHo Jang, Kyungwoo Song, Weonyoung Joo, Wanmo Kang, and Il-Chul Moon
Association for the Advancement of Artificial Intelligence (AAAI), 2021.
- [6] **Neutralizing Gender Bias in Word Embedding with Latent Disentanglement and Counterfactual Generation**
Seungjae Shin, Kyungwoo Song, Joonho Jang, Hyemi Kim, Weonyoung Joo, and Il-Chul Moon
Findings of Empirical Methods in Natural Language Processing (EMNLP), 2020.
- [7] **Deep Generative Positive-Unlabeled Learning under Selection Bias**
ByeongHu Na, Hyemi Kim, Kyungwoo Song, Weonyoung Joo, Yoonyeong Kim, and Il-Chul Moon
Conference on Information and Knowledge Management (CIKM), 2020.

Teaching Assistantships

Year	Course, School (Mean overall quality rating out of 5.0, if available)
Fall 2024	Introduction to Probability and Statistics, Columbia IEOR (4.67)
Spring 2024	Stochastic Models, Columbia IEOR (4.64)
Fall 2023	Analytics in Action, Columbia Business School
Spring 2023	Transportation Analytics & Logistics, Columbia IEOR (4.6)
Spring 2022	Causal Inference, Columbia IEOR (4.4)
Fall 2018/19	Data Structure, KAIST ISysE

Honors and Awards

2025	Columbia-Dream Sports AI Innovation Fellowship, Columbia University
2024	Finalist, INFORMS DEI Best Student Paper Award
2024	Finalist, INFORMS Transportation Science and Logistics Society Best Student Paper Award
2023	Finalist, INFORMS 2023 BSS Data Challenge Competition (Report: Decoupling Price and Demand Estimation for Profit Optimization)
2023	Deming Doctoral Fellowship, Columbia University
2019	Songhyun Award, KAIST
2012	Presidential Science Scholarship, KFAS

Academic Service

2026	Session Chair, INFORMS Annual Meeting
2026	Session Chair, Fairness Accountability and Transparency (FAccT)
2026	Reviewer, Fairness Accountability and Transparency (FAccT)
2025 – 2026	Program Committee, IJCAI – Special Track on AI and Social Good
2024 – 2026	Program Committee, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)
	Reviewer, Socio-Economic Planning Sciences
	Reviewer, Journal of Risk and Insurance
2014	President, Student Council, Department of Industrial Engineering, KAIST

Industry Experience

Jun. 2024 –	Data Scientist Intern , <i>MOLOCO, Inc.</i> , Redwood City, USA
Aug. 2024	Designed and executed budget pacing simulations for online advertisements.
Jan. 2021 –	AI Research Intern , <i>NAVER Corporation</i> , Seongnam, Korea
Jul. 2021	Studied test-time training methods for robust machine learning.
Jun. 2015 –	Intern , <i>Samsung Electronics</i> , Suwon, Korea
Aug. 2015	Analyzed smart car development trends and improved patient analysis algorithms.

Talks

Fair Aggregation in Virtual Power Plants

- 2026 INFORMS RMP Conference, Ann Arbor, Michigan
- 2026 ACM FAccT Conference, Montreal, Canada
- 2026 Market Innovation Workshop, Virtual

Learning Fair Demand Models

- 2025 INFORMS Annual Meeting, Atlanta, Georgia
- 2025 INFORMS RMP Conference, New York, NY

Fair Fares for Vehicle Sharing Systems

- 2025 ACM FAccT Conference, Athena, Greek
- 2025 Seminar at KAIST, Daejeon, Korea
- 2025 INFORMS TSL Workshop, Seoul, Korea
- 2024 INFORMS DEI Best Student Paper Awards, Seattle, WA
- 2024 INFORMS TSL Best Student Paper Awards, Seattle, WA
- 2024 INFORMS RMP Conference, Los Angeles, CA
- 2024 Market Innovation Workshop, Virtual
- 2023 INFORMS Annual Meeting, Phoenix, AZ
- 2023 Workshop on AI & Analytics for Social Good, College Park, MD

Skills & Personal

- Programming Python, MATLAB, C; PyTorch, TensorFlow, Gurobi, Pyomo
- Languages English (advanced), Korean (native)
- Hobby Brazilian jiu-jitsu (purple belt)